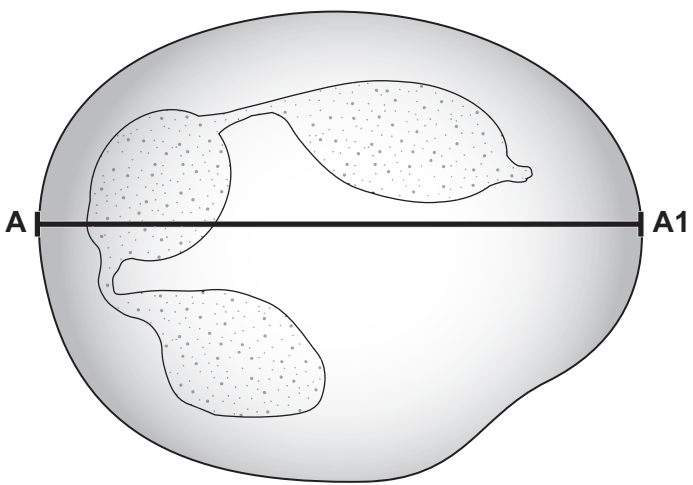


5. (a) Sarah was viewing a smear of blood under a light microscope. How could she calculate the magnification of the microscope she was using? [1]

.....

.....

(b) Sarah produced the drawing below. It shows one of the blood cells she looked at under the microscope.



(i) Name the type of white blood cell shown in the drawing above. [1]

.....

(ii) The actual size of the white blood cell, across the line **A – A1**, was 20 μm (20 micrometres). There are 1000 μm in 1 mm.

I. Measure the length of the line **A – A1** in the diagram above. [1]

length of the line **A – A1** = mm

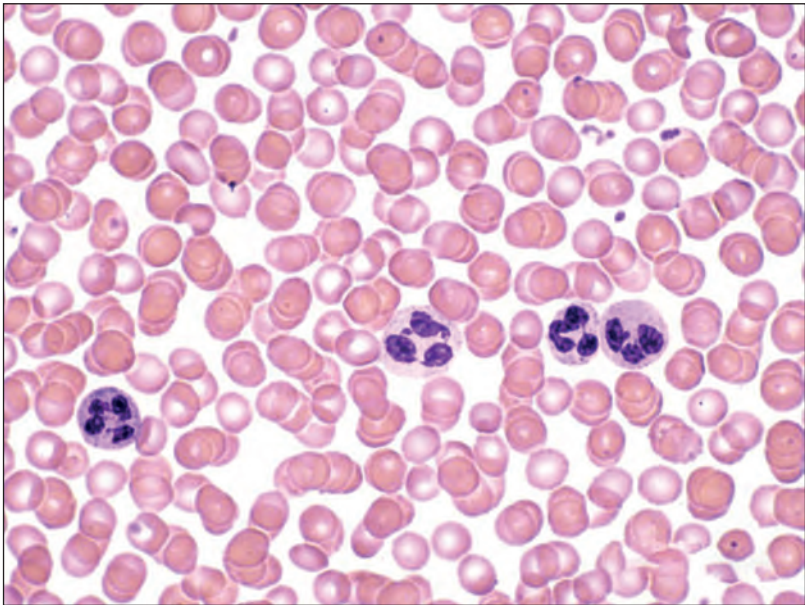
II. Calculate how many times bigger Sarah's drawing is than the actual blood cell. [2]

Magnification = $\times$



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only

- (c) The photograph below shows a smear of blood viewed under the high power magnification of a microscope.



- (i) Describe the procedure involved in preparing these blood cells for viewing under the microscope and state the reason why this is needed. [2]

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.....

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- (ii) The ratio of red blood cells to white blood cells is 500:1. A sample of blood has 57 000 red blood cells.
Calculate the number of white blood cells in the sample. [1]

Number of white blood cells =

8

